Linear Algebra — Exam Paper

1. Compute the determinant

$$\begin{vmatrix} a_1+\lambda & a_2 & a_3 & \cdots & a_n \\ a_1 & a_2+\lambda & a_3 & \cdots & a_n \\ a_1 & a_2 & a_3+\lambda & \cdots & a_n \\ \vdots & \vdots & \vdots & & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_n+\lambda \end{vmatrix}$$

2. Let $\alpha = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$. Find. $\alpha^{\mathrm{T}}\alpha,\ \alpha\alpha^{\mathrm{T}},\ \alpha\alpha^{\mathrm{T}}\alpha,\ (\alpha\alpha^{\mathrm{T}})^{100}$

3. Let A be a square matrix of order n satisfying $A^2 - 5A + E = 0$; prove that $A - E$ is invertible, and find its inverse.

4. Discuss for which values of a the following system: (1) has a unique solution; (2) has no solution; (3) has infinitely many solutions? And when it has infinitely many solutions, find the general solution,

$$\begin{cases} ax_1 + x_2 + x_3 = 0 \\ x_1 + ax_2 + x_3 = 2 \\ x_1 + x_2 + ax_3 = a \end{cases}$$

5. Let α_1, α_2, α_3 be linearly independent, and let $\beta_1 = \alpha_1 + \alpha_2$, $\beta_2 = \alpha_2 + \alpha_3$, $\beta_3 = \alpha_3 + \alpha_1$. Prove that β_1, β_2, β_3 are linearly independent.

6. Determine a, b such that $A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & 2 & 1 & -3 & a \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & b \end{pmatrix}$ has rank 2; then find a maximal linearly independent subset of the set of column vectors of A, and express the remaining vectors in terms of the maximal linearly independent subset.

7. Let $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ or $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$; find the eigenvalues and eigenvectors of A